Contact

www.linkedin.com/in/heath-mccabe (LinkedIn)

Top Skills

Conan
CMake
C++

Certifications

Amazon Web Services Solutions Architect - Associate

Honors-Awards

Dean’s List (x7)
President’s Gold Scholarship
Eagle Scout
NexGen Awards (x10 in total)

Publications

Tunable, Hardware-based Quantum Random Number Generation using Coupled Quantum Dots

Heath McCabe

Senior Software Engineer at Shield AI | MSECE at Johns Hopkins
Washington DC-Baltimore Area

Summary

Software engineer experienced with developing solutions to business problems ranging from RF signal modeling and analysis to developing microservices in AWS or within edge devices.

I hold a Master of Science from John’s Hopkins and a Bachelor of science from Baylor University. I am excited by all things tech and passionate about developing high-quality software to drive solutions for engineering and business problems. As a detail-oriented and results-driven professional, I am excited to leverage my skills to contribute to the growth and success of a dynamic organization.

Experience

Shield AI
Senior Software Engineer
December 2025 - Present (7 months)
Arlington, Virginia, United States

Building the world’s best AI pilot

Innovative Defense Technologies (IDT)
Software Engineer I
March 2025 - December 2025 (10 months)
Arlington, VA

Building software solutions to challenging problems in the defense space in IDT’s Advanced Capabilities group

Capital One
Software Engineer
June 2022 - March 2025 (2 years 10 months)
Arlington, Virginia, United States

Building resilient, distributed back-end services using Java and Spring Boot to empower our Auto Finance business.

Lockheed Martin

3 years 1 month

Guidance Navigation & Control Engineer
March 2022 - June 2022 (4 months)
Dallas, Texas, United States

• Navigation Team Member (∈ GNC Team)
• Hardware-In-The-Loop lab Resident-In-Charge

• Led planning phase of HWIL Verification and Validation effort

Systems Engineer Associate
June 2019 - March 2022 (2 years 10 months)
Dallas/Fort Worth Area

• Navigation Team Member (∈ GNC Team)
• Hardware-In-The-Loop lab Team

• Built C++ interface for integrating GPS Satellite Simulator into Hardware-In-The-Loop Lab to enable testing more advanced GPS capabilities
• Wrote C++ utility to convert hex GPS Navigation data dump into usable format for GPS hot start
• Wrote concurrent Python code to implement batch-running of Hardware-In-The-Loop (HWIL) scenarios which resulted in time savings when performing large amounts of HWIL testing at once
• Performed RF link analysis in Matlab to predict GPS performance in flight
• Collaborated with software team to verify missile software meets requirements via HWIL testing
• Performed GPS & inertial navigation analysis for development missile program which resulted in greater confidence in GPS capabilities and presented findings to program and senior managers
• Led execution of testing GPS antennas and filters in lab and live-sky tests to verify requirements are met
• Performed hundreds of HWIL runs on development and flight hardware which provides confidence in the flight test hardware, aids in GNC algorithm development, and supports flight software development
• Created procedure for testing missile launch pod RF path and executed this test ahead of flight tests alongside members of Test Engineering and Electrical Engineering teams

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• Received formal recognition from Director, Chief Engineer, E&T Manager, and peers for quality work and dedication

General Dynamics Mission Systems

Electrical Engineering Intern

May 2018 - August 2018 (4 months)

Plano, TX

• Electrical Hardware Engineering Team

• Developed an automation tool in Python to merge data from multiple sources to automate the quarterly part obsolescence reports for the hardware engineering team saving 10+ hours of work each quarter

• Performed WEEE (environmental compliance) test on engineering assembly

• Performed environmental temperature testing on COTS circuit board to verify that it could operate in the environments required for use in our system

Education

Johns Hopkins Whiting School of Engineering

Master of Science in Engineering, Electrical and Computer Engineering

Baylor University

Bachelor of Science in Engineering, Electrical and Electronics Engineering